

Rosalie Miller

WA Bee Atlas Collection and Identification Report - 2024

1 Your 2024 Collections

Rosalie Miller caught 43 bees across 5 counties from April 14, 2024 to July 11, 2024 representing 18 unique taxa, including 17 unique species.

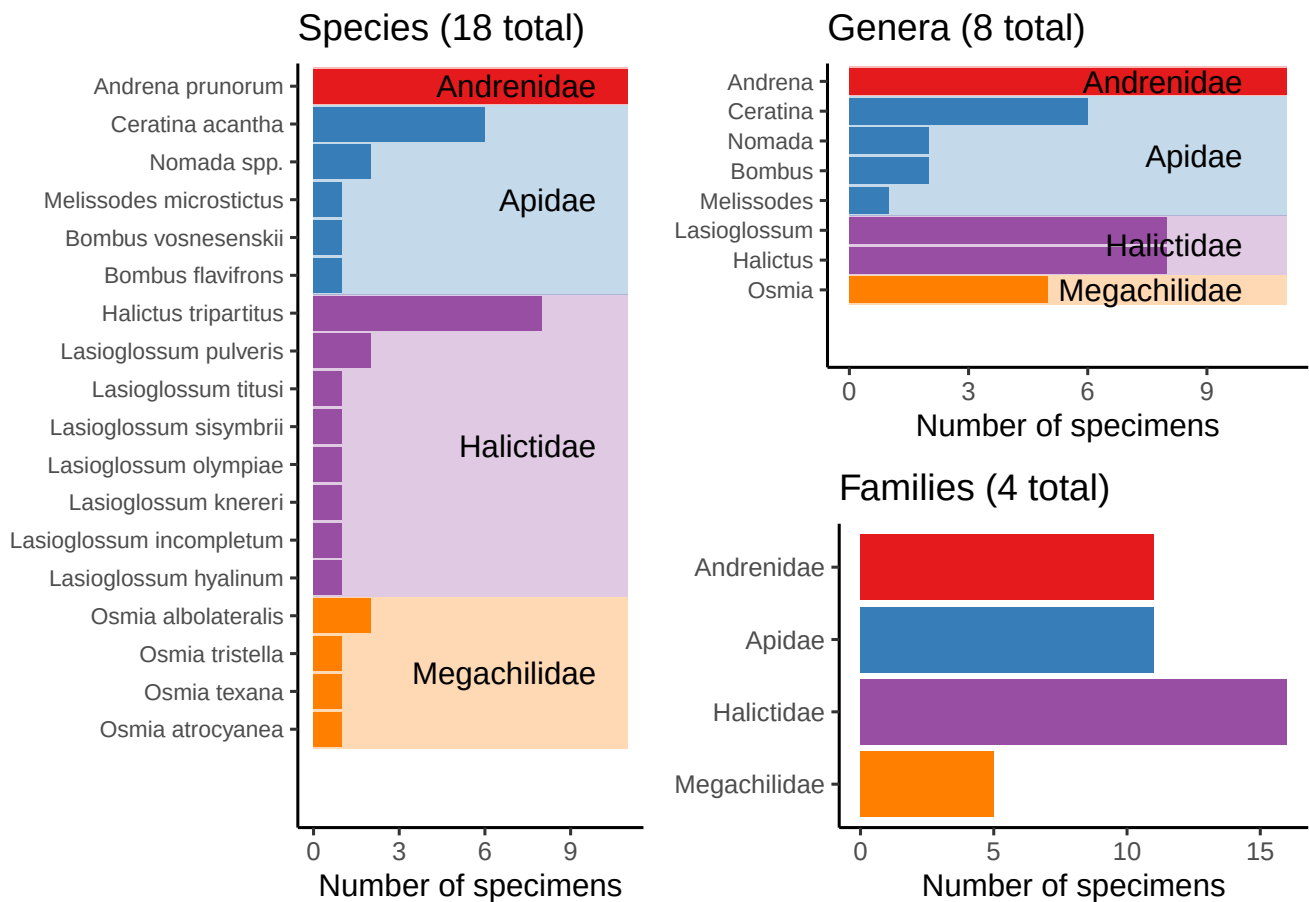


Figure 1: Bees caught by Rosalie Miller, broken down by species, genus, and family.

2 All Your Collections

Rosalie Miller caught 113 bees across 5 counties from April 14, 2024 to August 17, 2025, representing 35 unique taxa, including 27 unique species.

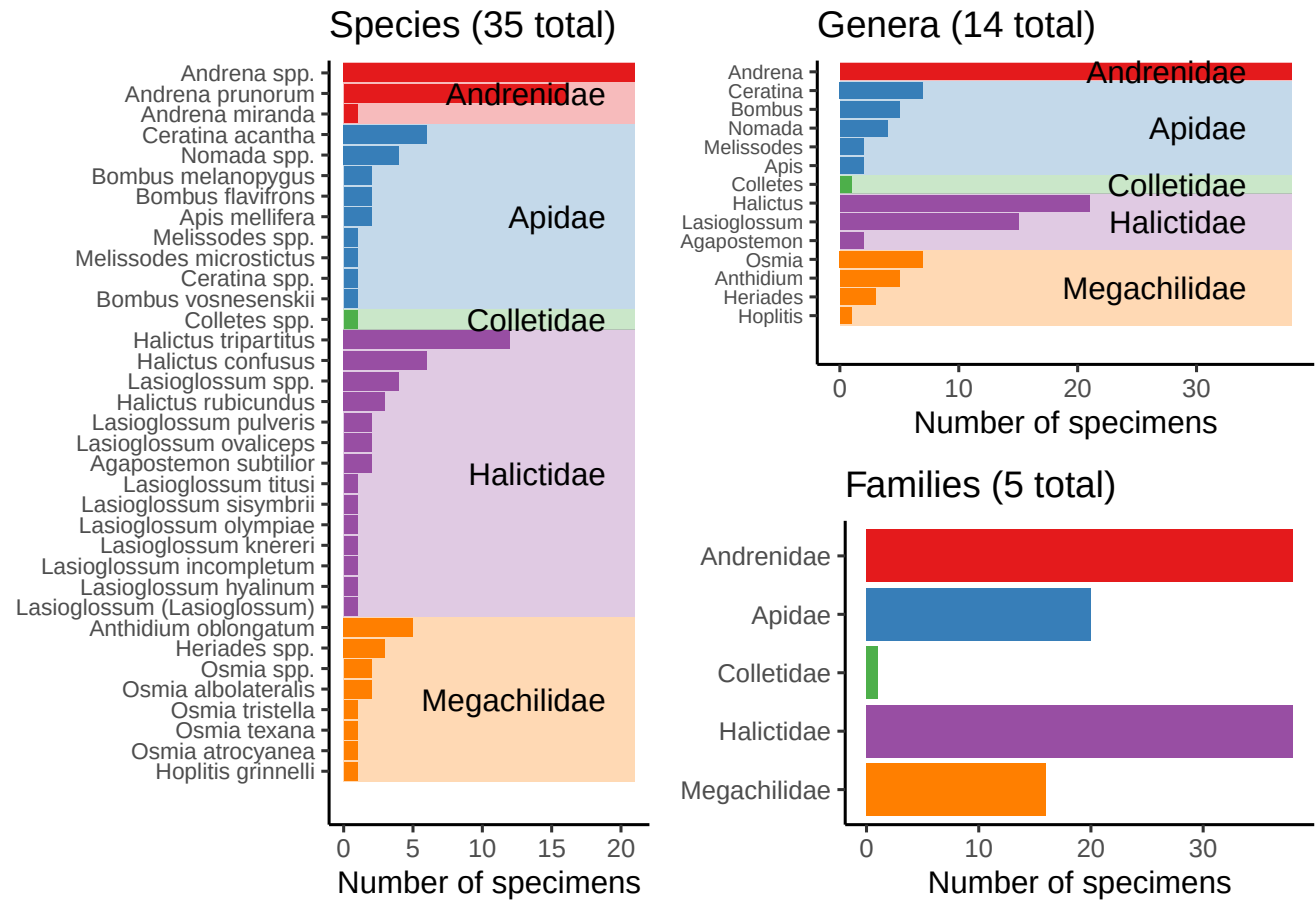


Figure 2: Bees caught by Rosalie Miller, broken down by species, genus, and family.

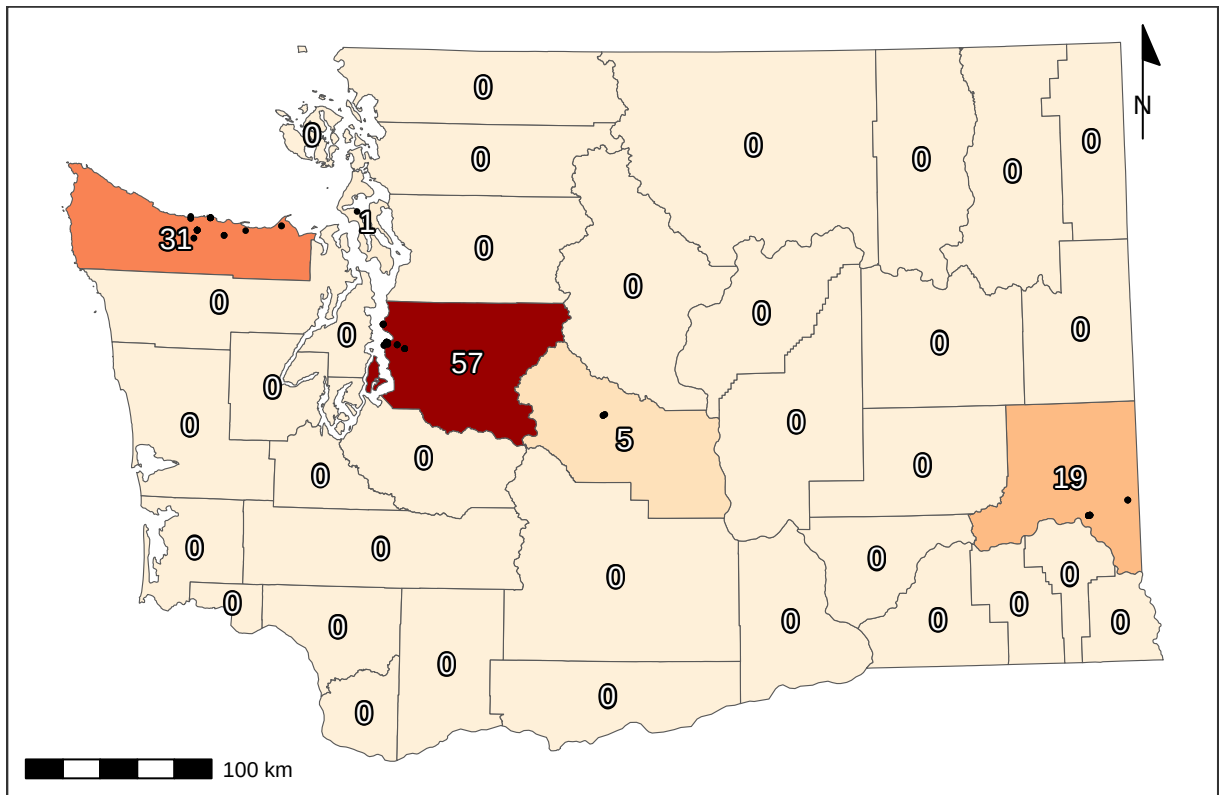


Figure 3: Bee catch locations for Rosalie Miller (within Washington), along with total catches per county.

3 Total Catches

Volunteers from the Washington Bee Atlas project caught 10006 bees across 35 counties from March 10, 2024 to November 14, 2024, representing 354 unique species and 43 unique genera. The **Nimble Net Kudos** (most specimens collected) goes to Karla Salp, Karen Wright, and Ingrid Carmean, who caught a total of 1689, 1005, and 957 specimens. The *positive* kind of **Darwin Award** (most species collected) goes to Karla Salp, Karen Wright, and Ellery Newcomer, who caught a total of 220, 167, and 153 unique species. Well done!

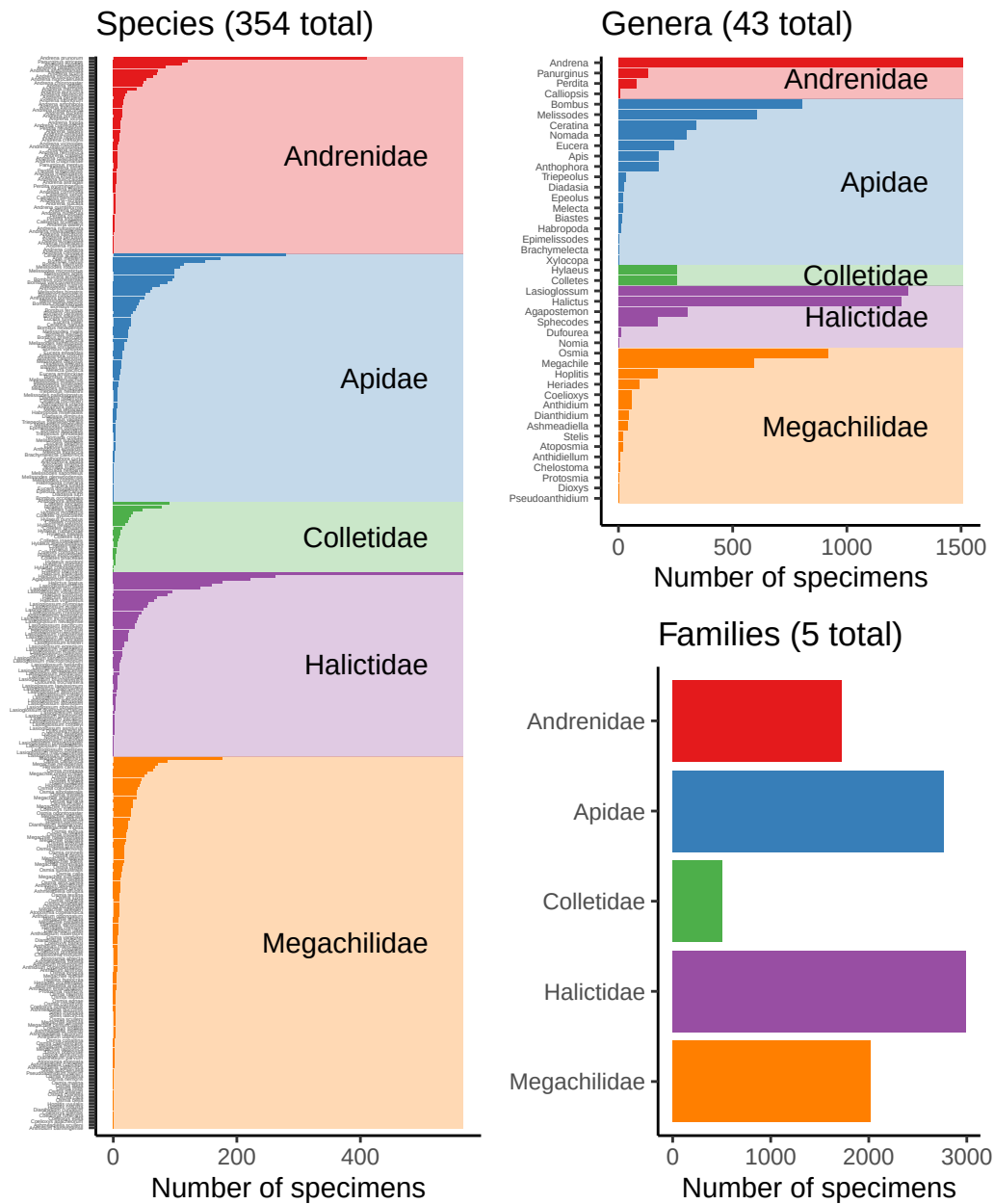


Figure 4: Bees caught by all volunteers, broken down by species, genus, and family.

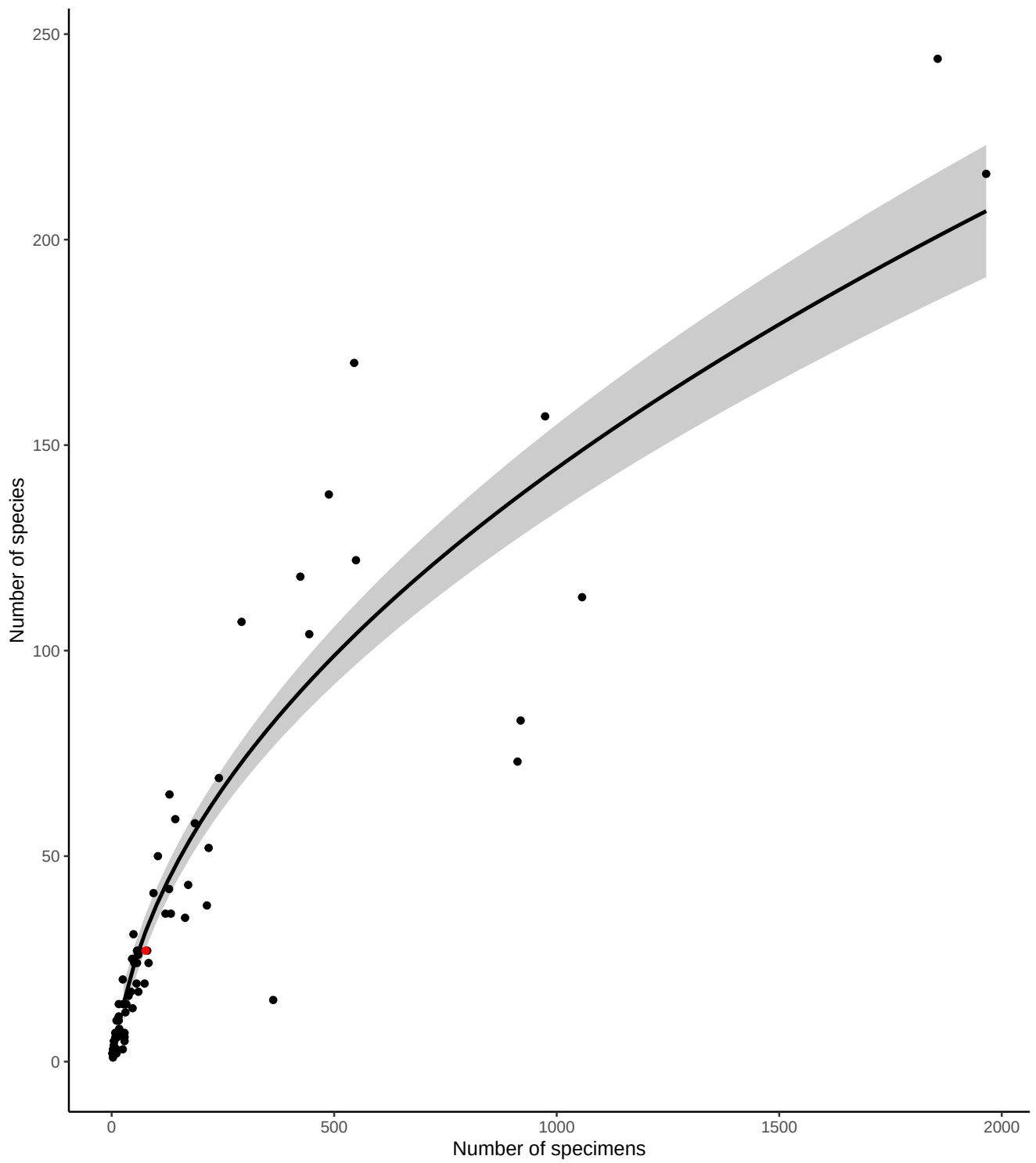


Figure 5: Number of bee specimens and unique bee species caught by all volunteers, with your effort shown in red. This graph should give you an idea of how many specimens you would need to catch to begin seeing rarer bee species.



Figure 6: Total specimens caught per county, along with catch location of each specimen (black dots). For genus- and species-specific information for each county, see Tables 3 and 4.

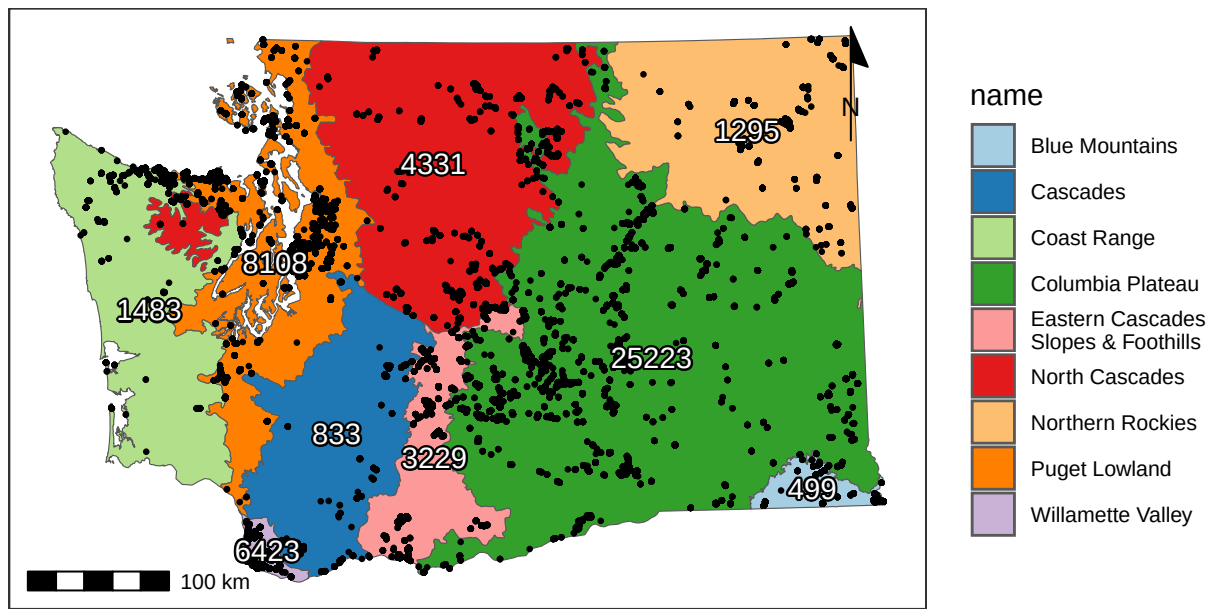


Figure 7: Total catches per ecoregion (Level III), along with catch location of each specimen (black dots).

4 Flight Phenology

4.1 West of (and including) the Cascade Mountains

Most bees (90%) were caught between April 16 and September 01, but the peak of season (50% of specimens) was from June 11 to July 27.

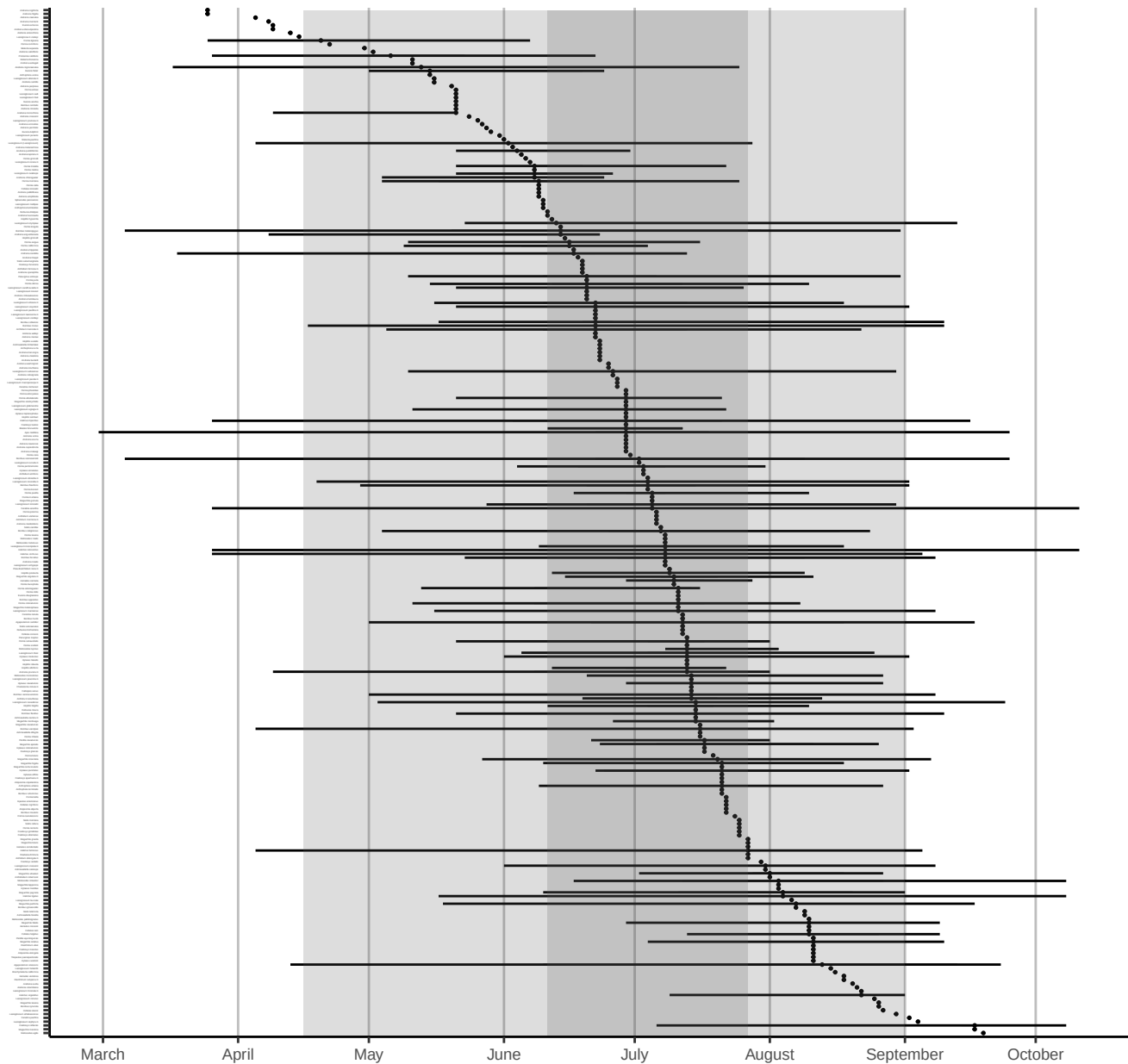


Figure 8: Phenology plot for all bee species caught in or West of the Cascade Mountains, sorted by median abundance times. Percentiles of overall emergence times (50th & 90th) are shown in grey shaded regions. Date ranges for each species (minimum, first, second, third quartiles, and maximum) are shown only for species with >10 specimens.

4.2 East of the Cascade Mountains

Most bees (90%) were caught between April 14 and September 18, but the peak of season (50% of specimens) was from May 15 to July 16.



Figure 9: Phenology plot for all bee species caught east of the Cascade Mountains, sorted by median abundance times. Percentiles of overall emergence times (50th & 90th) are shown in grey shaded regions. Date ranges for each species (minimum, first, second, third quartiles, and maximum) are shown only for species with >10 specimens.

5 Plant genera

Volunteers collected specimens from a total of 371 unique flower genera, with most volunteers sampling from 12 flower genera (median value). The **Flower Power Kudos** (most sampled flower genera) goes to Karla Salp, Karen Wright, and Nancy Carlson, who collected bees from a total of 170, 155, and 118 genera of flowers. Well done!

The flower genera that had the most specimens caught on them were *Grindelia*, *Ericameria*, and *Lomatium*, which yielded a total of 1095, 731, and 717 specimens. The flower genera that were popular with the most species of bees were *Erigeron*, *Phacelia*, and *Cirsium*, hosting a total of 101, 98, and 91 unique bee species. See Tables 1 and 2 for more details.



Figure 10: Recorded number of flower genera per county.

Table 1: Number of bee specimens collected from each plant genus. Plants with few records are great targets for future sampling.

Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count
<i>Grindelia</i>	1095	<i>Amsinckia</i>	97	<i>Delphinium</i>	35	<i>Angelica</i>	18	<i>Armeria</i>	8	<i>Anthemis</i>	4	<i>Abelia</i>	2
<i>Ericameria</i>	731	<i>Berberis</i>	97	<i>Malva</i>	35	<i>Acmispon</i>	17	<i>Coriandrum</i>	8	<i>Artemisia</i>	3	<i>Androsace</i>	2
<i>Lomatium</i>	717	<i>Dieteria</i>	94	<i>Nemophila</i>	34	<i>Euphorbia</i>	17	<i>Cuphea</i>	8	<i>Asparagus</i>	3	<i>Arenaria</i>	2
<i>Hypochaeris</i>	655	<i>Eschscholzia</i>	94	<i>Syringa</i>	34	<i>Cacaliopsis</i>	16	<i>Cytisus</i>	8	<i>Barbarea</i>	3	<i>Cardamine</i>	2
<i>Phacelia</i>	624	<i>Fragaria</i>	93	<i>Drymocalis</i>	33	<i>Lewisia</i>	16	<i>Dipsacus</i>	8	<i>Dimorphotheca</i>	3	<i>Corydalis</i>	2
<i>Helianthus</i>	490	<i>Origanum</i>	91	<i>Descurainia</i>	32	<i>Campanula</i>	15	<i>Downingia</i>	8	<i>Eremogone</i>	3	<i>Cota</i>	2
<i>Balsamorhiza</i>	471	<i>Clarkia</i>	89	<i>Malus</i>	32	<i>Draba</i>	15	<i>Hesperochiron</i>	8	<i>Galium</i>	3	<i>Crocasmia</i>	2
<i>Cirsium</i>	450	<i>Medicago</i>	88	<i>Crataegus</i>	31	<i>Hackelia</i>	15	<i>Aquilegia</i>	7	<i>Gayophytum</i>	3	<i>Crocus</i>	2
<i>Erigeron</i>	446	<i>Ceanothus</i>	85	<i>Doellingeria</i>	31	<i>Heterotheca</i>	15	<i>Erica</i>	7	<i>Ceanothus</i>	3	<i>Achlys</i>	1
<i>Solidago</i>	429	<i>Daucus</i>	83	<i>Myosotis</i>	31	<i>Lonicera</i>	15	<i>Fritillaria</i>	7	<i>Hibiscus</i>	3	<i>Adelphia</i>	1
<i>Penstemon</i>	414	<i>Sphaeralcea</i>	82	<i>Tunacetum</i>	31	<i>Oenothera</i>	15	<i>Viburnum</i>	7	<i>Impatiens</i>	3	<i>Albizia</i>	1
<i>Leucanthemum</i>	413	<i>Lathyrus</i>	78	<i>Bellis</i>	30	<i>Perideridia</i>	15	<i>Amelanchier</i>	6	<i>Lapsana</i>	3	<i>Anthozanthum</i>	1
<i>Sisymbrium</i>	408	<i>Pediocactus</i>	78	<i>Cleome</i>	30	<i>Ratibida</i>	15	<i>Anchusa</i>	6	<i>Liatris</i>	3	<i>Antirrhinum</i>	1
<i>Eriogonum</i>	381	<i>Epilobium</i>	77	<i>Hieracium</i>	30	<i>Antennaria</i>	14	<i>Cynara</i>	6	<i>Lobularia</i>	3	<i>Bellardia</i>	1
<i>Lupinus</i>	372	<i>Rudbeckia</i>	77	<i>Rorippa</i>	30	<i>Crocidium</i>	14	<i>Cynoglossum</i>	6	<i>Nothochelone</i>	3	<i>Boechera</i>	1
<i>Potentilla</i>	320	<i>Monardella</i>	74	<i>Cerastium</i>	29	<i>Frangula</i>	14	<i>Limnanthes</i>	6	<i>Oenanthe</i>	3	<i>Buglossoides</i>	1
<i>Rubus</i>	285	<i>Physocarpus</i>	71	<i>Collinsia</i>	29	<i>Hydrophyllum</i>	14	<i>Luetkea</i>	6	<i>Oreocarya</i>	3	<i>Calystegia</i>	1
<i>Rosa</i>	269	<i>Camassia</i>	70	<i>Lactuca</i>	29	<i>Plantago</i>	14	<i>Microseris</i>	6	<i>Physaria</i>	3	<i>Capsella</i>	1
<i>Symphoricarpos</i>	265	<i>Spiraea</i>	70	<i>Olaynia</i>	29	<i>Borago</i>	13	<i>Nigella</i>	6	<i>Pieris</i>	3	<i>Carduus</i>	1
<i>Taraxacum</i>	258	<i>Calochortus</i>	69	<i>Clematis</i>	28	<i>Heleum</i>	13	<i>Onopordum</i>	6	<i>Primula</i>	3	<i>Cercis</i>	1
<i>Madia</i>	256	<i>Nepeta</i>	68	<i>Acer</i>	27	<i>Bistorta</i>	12	<i>Phoenixaulis</i>	6	<i>Pseudognaphalium</i>	3	<i>Chamaecrista</i>	1
<i>Holodiscus</i>	247	<i>Coreopsis</i>	67	<i>Canadanthus</i>	27	<i>Cucurbita</i>	12	<i>Robinia</i>	6	<i>Sagittaria</i>	3	<i>Comandra</i>	1
<i>Achillea</i>	246	<i>Agastache</i>	66	<i>Sonchus</i>	27	<i>Heuchera</i>	12	<i>Ageratina</i>	5	<i>Saponaria</i>	3	<i>Cotoneaster</i>	1
<i>Asclepias</i>	228	<i>Hypericum</i>	60	<i>Sphaerophysa</i>	26	<i>Persicaria</i>	12	<i>Ajuga</i>	5	<i>Silene</i>	3	<i>Cryptantha</i>	1
<i>Lotus</i>	224	<i>Convolvulus</i>	59	<i>Linum</i>	25	<i>Spergularia</i>	12	<i>Ammi</i>	5	<i>Tellima</i>	3	<i>Cucumis</i>	1
<i>Trifolium</i>	218	<i>Cichorium</i>	55	<i>Wyethia</i>	25	<i>Vitex</i>	12	<i>Arctostaphylos</i>	5	<i>Thalictrum</i>	3	<i>Dicentra</i>	1
<i>Crepis</i>	217	<i>Veronica</i>	55	<i>Aruncus</i>	24	<i>Centaurium</i>	11	<i>Bassia</i>	5	<i>Thermopsis</i>	3	<i>Echinops</i>	1
<i>Brassica</i>	213	<i>Arnica</i>	54	<i>Chondrilla</i>	24	<i>Chorispura</i>	11	<i>Cymopterus</i>	5	<i>Visnaga</i>	3	<i>Erythronium</i>	1
<i>Centaurea</i>	210	<i>Digitalis</i>	54	<i>Lithospermum</i>	24	<i>Cosmos</i>	11	<i>Helopsis</i>	5	<i>Dalea</i>	2	<i>Gnaphalium</i>	1
<i>Prunus</i>	202	<i>Linaria</i>	53	<i>Anethum</i>	23	<i>Dahlia</i>	11	<i>Hydrangea</i>	5	<i>Eurybia</i>	2	<i>Holcus</i>	1
<i>Salix</i>	200	<i>Chaenactis</i>	52	<i>Purshia</i>	23	<i>Micranthes</i>	11	<i>Melissa</i>	5	<i>Fagopyrum</i>	2	<i>Hylotelephium</i>	1
<i>Ribes</i>	188	<i>Nothocalais</i>	51	<i>Hieracium</i>	22	<i>Pseudotsuga</i>	11	<i>Mycelis</i>	5	<i>Fothergilla</i>	2	<i>Hymenopappus</i>	1
<i>Ranunculus</i>	184	<i>Sedum</i>	51	<i>Mentzelia</i>	22	<i>Pyrocampa</i>	11	<i>Narcissus</i>	5	<i>Hesperis</i>	2	<i>Hyssopus</i>	1
<i>Anaphalis</i>	179	<i>Euthamia</i>	47	<i>Thelypodium</i>	22	<i>Valerianella</i>	11	<i>Petasites</i>	5	<i>Ionactis</i>	2	<i>Ilex</i>	1
<i>Senecio</i>	176	<i>Cornus</i>	44	<i>Triteleia</i>	22	<i>Asperugo</i>	10	<i>Phedimus</i>	5	<i>Kalmia</i>	2	<i>Ipomopsis</i>	1
<i>Chrysothamnus</i>	175	<i>Helianthella</i>	44	<i>Papaver</i>	21	<i>Berteroa</i>	10	<i>Sorbus</i>	5	<i>Ladcania</i>	2	<i>Isocoma</i>	1
<i>Melilotus</i>	167	<i>Lavandula</i>	44	<i>Polemonium</i>	21	<i>Erodium</i>	10	<i>Argentina</i>	4	<i>Ligusticum</i>	2	<i>Lamprocapnos</i>	1
<i>Gaillardia</i>	154	<i>Mentha</i>	44	<i>Erythranthe</i>	20	<i>Lamium</i>	10	<i>Cistus</i>	4	<i>Lycium</i>	2	<i>Lomelosia</i>	1
<i>Chamaenerion</i>	147	<i>Plagiobothrys</i>	44	<i>Frasera</i>	20	<i>Lygodesmia</i>	10	<i>Dichelostemma</i>	4	<i>Mertensia</i>	2	<i>Lycopus</i>	1
<i>Philadelphus</i>	146	<i>Plectritis</i>	43	<i>Iris</i>	20	<i>Marrubium</i>	10	<i>Echinacea</i>	4	<i>Montia</i>	2	<i>Matricaria</i>	1
<i>Allium</i>	144	<i>Pastinaca</i>	42	<i>Oenleria</i>	20	<i>Monarda</i>	10	<i>Elaeagnus</i>	4	<i>Nasturtium</i>	2	<i>Muscari</i>	1
<i>Lepidium</i>	134	<i>Rhododendron</i>	42	<i>Phlox</i>	20	<i>Solanum</i>	10	<i>Eutrochium</i>	4	<i>Noccea</i>	2	<i>Nicotiana</i>	1
<i>Symphoricarpos</i>	132	<i>Vaccinium</i>	42	<i>Rhus</i>	20	<i>Tragopogon</i>	10	<i>Glycyrrhiza</i>	4	<i>Osmia</i>	2	<i>Oxalis</i>	1
<i>Vicia</i>	120	<i>Lithophragma</i>	40	<i>Toxicoscordion</i>	20	<i>Fallopia</i>	9	<i>Iberis</i>	4	<i>Ozomelis</i>	2	<i>Phalaris</i>	1
<i>Cilia</i>	119	<i>Erysimum</i>	39	<i>Abronia</i>	19	<i>Pyrrus</i>	9	<i>Ivesia</i>	4	<i>Pedicularis</i>	2	<i>Pisum</i>	1
<i>Salvia</i>	117	<i>Nestodus</i>	39	<i>Anthriscus</i>	19	<i>Rhaphiolepis</i>	9	<i>Machaeranthera</i>	4	<i>Petroselinum</i>	2	<i>Polygonum</i>	1
<i>Jacobaea</i>	116	<i>Calendula</i>	38	<i>Castilleja</i>	19	<i>Tripleurospermum</i>	9	<i>Microsteris</i>	4	<i>Pilosella</i>	2	<i>Salsola</i>	1
<i>Eriophyllum</i>	111	<i>Dasiphora</i>	38	<i>Gaultheria</i>	19	<i>Leontodon</i>	8	<i>Mutarda</i>	4	<i>Poa</i>	2	<i>Sambucus</i>	1
<i>Astragalus</i>	104	<i>Sidalcea</i>	38	<i>Thymus</i>	19	<i>Levistica</i>	8	<i>Opuntia</i>	4	<i>Pyraecantha</i>	2	<i>Santolina</i>	1
<i>Apocynum</i>	102	<i>Cakile</i>	37	<i>Collomia</i>	18	<i>Paeonia</i>	8	<i>Petrosedum</i>	4	<i>Satureja</i>	2	<i>Sisyrinchium</i>	1
<i>Geranium</i>	99	<i>Claytonia</i>	37	<i>Oxytropis</i>	18	<i>Scrophularia</i>	8	<i>Stellaria</i>	4	<i>Symphytum</i>	2	<i>Urtica</i>	1
<i>Ilamna</i>	99	<i>Prunella</i>	35	<i>Stephanomeria</i>	18	<i>Stachys</i>	8	<i>Tolmiea</i>	4	<i>Viola</i>	2	<i>Veratrum</i>	1
<i>Lythrum</i>	97	<i>Rhaponticum</i>	35	<i>Tagetes</i>	18	<i>Valeriana</i>	8	<i>Verbena</i>	4	<i>Zinnia</i>	2	<i>Xerophyllum</i>	1

Table 2: Number of bee species collected from each plant genus

Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count
<i>Erigeron</i>	101	<i>Coreopsis</i>	31	<i>Bellis</i>	15	<i>Antennaria</i>	8	<i>Ageratina</i>	4	<i>Argentina</i>	2	<i>Abelia</i>	1
<i>Phacelia</i>	98	<i>Agastache</i>	29	<i>Crataegus</i>	14	<i>Aruncus</i>	8	<i>Ajuga</i>	4	<i>Artemisia</i>	2	<i>Achlys</i>	1
<i>Cirsium</i>	91	<i>Eschscholzia</i>	29	<i>Hieracium</i>	14	<i>Berteroa</i>	8	<i>Ammi</i>	4	<i>Borago</i>	2	<i>Adelphia</i>	1
<i>Grindelia</i>	91	<i>Monardella</i>	29	<i>Hieracium</i>	14	<i>Cacaliopsis</i>	8	<i>Cistus</i>	4	<i>Cardamine</i>	2	<i>Albizia</i>	1
<i>Eriogonum</i>	86	<i>Berberis</i>	28	<i>Malus</i>	14	<i>Canadanthus</i>	8	<i>Cosmos</i>	4	<i>Cota</i>	2	<i>Androsace</i>	1
<i>Penstemon</i>	85	<i>Chrysothamnus</i>	28	<i>Malva</i>	14	<i>Cleome</i>	8	<i>Cuphea</i>	4	<i>Crocosmia</i>	2	<i>Anthozanthum</i>	1
<i>Solidago</i>	82	<i>Convolvulus</i>	28	<i>Tanacetum</i>	14	<i>Erysimum</i>	8	<i>Cymopterus</i>	4	<i>Dalea</i>	2	<i>Antirrhinum</i>	1
<i>Crepis</i>	75	<i>Dieteria</i>	28	<i>Lithospermum</i>	13	<i>Heuchera</i>	8	<i>Dahlia</i>	4	<i>Eurybia</i>	2	<i>Arenaria</i>	1
<i>Hypochaeris</i>	75	<i>Fragaria</i>	28	<i>Parshia</i>	13	<i>Hydrophyllum</i>	8	<i>Dichlostenma</i>	4	<i>Galium</i>	2	<i>Bellardia</i>	1
<i>Lupinus</i>	75	<i>Calochortus</i>	27	<i>Sonchus</i>	13	<i>Anethum</i>	7	<i>Echinacea</i>	4	<i>Geum</i>	2	<i>Boechera</i>	1
<i>Leucanthemum</i>	69	<i>Chaenactis</i>	27	<i>Collomia</i>	12	<i>Anthriscus</i>	7	<i>Eutrochium</i>	4	<i>Glycyrrhiza</i>	2	<i>Buglossoides</i>	1
<i>Rosa</i>	69	<i>Lathyrus</i>	27	<i>Doellingeria</i>	12	<i>Campanula</i>	7	<i>Frangula</i>	4	<i>Hesperis</i>	2	<i>Calystegia</i>	1
<i>Potentilla</i>	67	<i>Madia</i>	26	<i>Hackelia</i>	12	<i>Centaurium</i>	7	<i>Fritillaria</i>	4	<i>Iberis</i>	2	<i>Capsella</i>	1
<i>Sisymbrium</i>	64	<i>Medicago</i>	26	<i>Lactuca</i>	12	<i>Helenium</i>	7	<i>Heliopsis</i>	4	<i>Impatiens</i>	2	<i>Carduus</i>	1
<i>Achillea</i>	63	<i>Hypericum</i>	25	<i>Plantago</i>	12	<i>Heterotheca</i>	7	<i>Hydrangea</i>	4	<i>Ionactis</i>	2	<i>Cercis</i>	1
<i>Ericameria</i>	63	<i>Rudbeckia</i>	25	<i>Syringa</i>	12	<i>Micranthes</i>	7	<i>Ivesia</i>	4	<i>Kalmia</i>	2	<i>Chamaecrista</i>	1
<i>Lomatium</i>	63	<i>Arnica</i>	24	<i>Triteleia</i>	12	<i>Perideridia</i>	7	<i>Machaeranthera</i>	4	<i>Levisticum</i>	2	<i>Comandra</i>	1
<i>Balsamorhiza</i>	62	<i>Lythrum</i>	24	<i>Vaccinium</i>	12	<i>Rorippa</i>	7	<i>Microsteris</i>	4	<i>Liatris</i>	2	<i>Corydalis</i>	1
<i>Rubus</i>	62	<i>Nepeta</i>	24	<i>Wyethia</i>	12	<i>Stephanomeria</i>	7	<i>Mycelis</i>	4	<i>Ligusticum</i>	2	<i>Cotoneaster</i>	1
<i>Centauria</i>	60	<i>Cichorium</i>	23	<i>Collinsia</i>	11	<i>Abronia</i>	6	<i>Narcissus</i>	4	<i>Limnanthes</i>	2	<i>Crocus</i>	1
<i>Chamaenerion</i>	59	<i>Clematis</i>	23	<i>Lonicera</i>	11	<i>Amelanchier</i>	6	<i>Oenleria</i>	4	<i>Labularia</i>	2	<i>Cryptantha</i>	1
<i>Allium</i>	58	<i>Epilobium</i>	23	<i>Mentzelia</i>	11	<i>Aquilegia</i>	6	<i>Onopordum</i>	4	<i>Lycium</i>	2	<i>Cucumis</i>	1
<i>Senecio</i>	51	<i>Claytonia</i>	22	<i>Olepnium</i>	11	<i>Armeria</i>	6	<i>Phedimus</i>	4	<i>Montia</i>	2	<i>Dicentra</i>	1
<i>Helianthus</i>	50	<i>Euthamia</i>	22	<i>Sphaerophysa</i>	11	<i>Castilleja</i>	6	<i>Stachys</i>	4	<i>Nasturtium</i>	2	<i>Downingia</i>	1
<i>Holodiscus</i>	50	<i>Ilama</i>	21	<i>Thelypodium</i>	11	<i>Crocidium</i>	6	<i>Toxicoscordion</i>	4	<i>Nigella</i>	2	<i>Echinops</i>	1
<i>Melilotus</i>	50	<i>Linaria</i>	21	<i>Thymus</i>	11	<i>Cucurbita</i>	6	<i>Anthemis</i>	3	<i>Nothochelone</i>	2	<i>Erythronium</i>	1
<i>Prunus</i>	50	<i>Nothocalais</i>	21	<i>Acer</i>	10	<i>Erica</i>	6	<i>Arctostaphylos</i>	3	<i>Opuntia</i>	2	<i>Fagopyrum</i>	1
<i>Trifolium</i>	50	<i>Sedum</i>	21	<i>Chondrilla</i>	10	<i>Fullopia</i>	6	<i>Asparagus</i>	3	<i>Osmia</i>	2	<i>Fothergilla</i>	1
<i>Lotus</i>	49	<i>Philadelphus</i>	20	<i>Descurainia</i>	10	<i>Hesperochiron</i>	6	<i>Barbarea</i>	3	<i>Oxytropis</i>	2	<i>Gnaphalium</i>	1
<i>Taraxacum</i>	49	<i>Physocarpus</i>	20	<i>Digitalis</i>	10	<i>Leontodon</i>	6	<i>Bassia</i>	3	<i>Pedicularis</i>	2	<i>Holcus</i>	1
<i>Anaphalis</i>	45	<i>Sphaeralcea</i>	20	<i>Euphorbia</i>	10	<i>Monarda</i>	6	<i>Cynara</i>	3	<i>Petroselinum</i>	2	<i>Hylotelephium</i>	1
<i>Gaillardia</i>	45	<i>Camassia</i>	19	<i>Lewisia</i>	10	<i>Persicaria</i>	6	<i>Dimorphotheca</i>	3	<i>Physaria</i>	2	<i>Hymenopappus</i>	1
<i>Salvia</i>	45	<i>Cornus</i>	19	<i>Oenothera</i>	10	<i>Rhaphiolepis</i>	6	<i>Elaeagnus</i>	3	<i>Pieris</i>	2	<i>Hyssopus</i>	1
<i>Brassica</i>	43	<i>Plectritis</i>	19	<i>Plagiobothrys</i>	10	<i>Spergularia</i>	6	<i>Eremogone</i>	3	<i>Pyraantha</i>	2	<i>Ilex</i>	1
<i>Ranunculus</i>	43	<i>Veronica</i>	19	<i>Polemonium</i>	10	<i>Tripleurospermum</i>	6	<i>Gayophytum</i>	3	<i>Stellaria</i>	2	<i>Ipomopsis</i>	1
<i>Symphotrichum</i>	43	<i>Daucus</i>	18	<i>Pseudotsuga</i>	10	<i>Valerianella</i>	6	<i>Hibiscus</i>	3	<i>Symphytum</i>	2	<i>Isocoma</i>	1
<i>Vicia</i>	43	<i>Delphinium</i>	18	<i>Acmispon</i>	9	<i>Anchusa</i>	5	<i>Lamium</i>	3	<i>Thermopsis</i>	2	<i>Ladecania</i>	1
<i>Lepidium</i>	42	<i>Drymocallis</i>	18	<i>Angelica</i>	9	<i>Asperugo</i>	5	<i>Lapsana</i>	3	<i>Viola</i>	2	<i>Lamprocapnos</i>	1
<i>Salix</i>	38	<i>Mentha</i>	18	<i>Bistorta</i>	9	<i>Coriandrum</i>	5	<i>Luetkea</i>	3	<i>Polygonum</i>	1	<i>Lomelosia</i>	1
<i>Symphoricarpos</i>	38	<i>Nastotus</i>	18	<i>Cerastium</i>	9	<i>Cynoglossum</i>	5	<i>Mutarda</i>	3	<i>Salsola</i>	1	<i>Lycopus</i>	1
<i>Asclepias</i>	37	<i>Prunella</i>	18	<i>Chorispora</i>	9	<i>Cytisus</i>	5	<i>Oenanthe</i>	3	<i>Sambucus</i>	1	<i>Matricaria</i>	1
<i>Amsinckia</i>	35	<i>Erythranthe</i>	17	<i>Erodium</i>	9	<i>Dipsacus</i>	5	<i>Oreocarya</i>	3	<i>Santolina</i>	1	<i>Melissa</i>	1
<i>Apocynum</i>	35	<i>Helianthella</i>	17	<i>Frasera</i>	9	<i>Draba</i>	5	<i>Phoenicautis</i>	3	<i>Satureja</i>	1	<i>Mertensia</i>	1
<i>Eriophyllum</i>	35	<i>Linum</i>	17	<i>Iris</i>	9	<i>Gaultheria</i>	5	<i>Primula</i>	3	<i>Sisyrinchium</i>	1	<i>Muscari</i>	1
<i>Geranium</i>	34	<i>Rhaponticum</i>	17	<i>Nemophila</i>	9	<i>Microseris</i>	5	<i>Pseudognaphalium</i>	3	<i>Tellima</i>	1	<i>Nicotiana</i>	1
<i>Jacobaea</i>	34	<i>Cakile</i>	16	<i>Pastinaca</i>	9	<i>Paeonia</i>	5	<i>Pyrus</i>	3	<i>Thalictrum</i>	1	<i>Noccea</i>	1
<i>Ribes</i>	34	<i>Pedocactus</i>	16	<i>Lithophragma</i>	8	<i>Petasites</i>	5	<i>Robinia</i>	3	<i>Tobmicea</i>	1	<i>Ozalis</i>	1
<i>Ceanothus</i>	33	<i>Sidalcea</i>	16	<i>Lygodesmia</i>	8	<i>Phlox</i>	5	<i>Sagittaria</i>	3	<i>Urtica</i>	1	<i>Ozomelis</i>	1
<i>Clarkia</i>	33	<i>Calendula</i>	15	<i>Marrubium</i>	8	<i>Pyrrocoma</i>	5	<i>Saponaria</i>	3	<i>Veratrum</i>	1	<i>Petrosedum</i>	1
<i>Origanum</i>	33	<i>Dasiphora</i>	15	<i>Papaver</i>	8	<i>Solanum</i>	5	<i>Scrophularia</i>	3	<i>Verbena</i>	1	<i>Phalaris</i>	1
<i>Spiraea</i>	33	<i>Lavandula</i>	15	<i>Ratibida</i>	8	<i>Tragopogon</i>	5	<i>Silene</i>	3	<i>Visnaga</i>	1	<i>Pilosella</i>	1
<i>Astragalus</i>	32	<i>Myosotis</i>	15	<i>Rhus</i>	8	<i>Valeriana</i>	4	<i>Sorbus</i>	3	<i>Xerophyllum</i>	1	<i>Pisum</i>	1
<i>Gilia</i>	32	<i>Rhododendron</i>	15	<i>Tagetes</i>	8	<i>Viburnum</i>	4	<i>Vitex</i>	3	<i>Zinnia</i>	1	<i>Poa</i>	1

6 County records

Table 3: Number of bee specimens from each county, by genus. You may want to focus your sampling in under-sampled counties.

	Adams	Asotin	Benton	Chelan	Columbia	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Island	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Shagit	Skamania	Snohomish	Spokane	Stevens	Thurston	Wahinkum	Walla Walla	Whatcom	Whitman	Yakima	TOTAL	
Agapostemon	2	0	3	0	34	194	3	2	30	1	2	4	110	0	3	11	15	0	16	1	0	1	0	11	0	0	0	2	1	0	6	2	0	17	0	1	1	20	10	503	
Andrena	6	15	25	71	201	305	13	2	90	5	3	25	341	0	1	37	142	5	700	118	11	71	2	67	0	4	0	1	40	16	5	57	30	29	0	29	13	361	392	3233	
Anthidiellum	0	0	0	0	0	1	0	0	0	0	3	0	8	0	0	0	0	0	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	20	
Anthidium	0	0	2	1	17	12	0	0	4	1	3	2	30	1	0	1	19	4	24	3	0	2	0	7	0	0	1	1	0	0	6	5	1	1	0	0	0	0	24	172	
Anthophora	2	1	9	6	36	3	0	0	8	0	10	0	84	0	2	25	5	0	26	9	0	2	0	16	0	1	0	0	1	0	0	4	0	0	0	4	5	9	43	311	
Apis	0	0	2	6	12	59	1	0	0	0	10	0	58	0	22	2	6	5	31	4	0	3	0	9	0	1	0	1	7	4	0	9	3	6	0	1	0	1	24	287	
Ashmeadiella	0	3	0	6	1	0	0	0	2	0	0	0	60	0	0	0	0	1	0	21	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	14	110	
Atoposmia	0	0	0	0	0	4	0	0	0	0	0	3	7	0	0	0	0	0	4	0	0	1	0	3	0	0	0	0	0	0	0	0	0	0	0	5	0	16	43		
Blastus	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14	
Bombus	1	5	7	33	297	128	7	1	18	4	7	14	95	9	13	304	19	105	239	29	0	8	5	68	0	43	2	1	33	7	6	15	22	40	1	20	14	24	230	1874	
Brachymelecta	0	1	1	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	3	9	
Calliopsis	0	0	0	0	0	0	0	0	3	0	0	0	35	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	44	
Ceratina	0	2	6	5	90	352	5	3	6	3	0	0	49	0	7	27	63	4	50	12	0	1	4	3	0	4	0	1	4	4	4	11	3	15	0	65	1	11	95	910	
Chelostoma	0	0	0	0	1	0	0	0	0	0	0	0	3	0	0	0	0	0	14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	22	
Coelioxys	1	3	3	0	24	21	0	1	0	0	4	0	32	0	0	0	10	0	5	0	0	0	0	0	0	0	1	1	0	0	1	0	0	3	0	1	0	0	10	121	
Colletes	0	8	15	2	168	7	0	0	12	0	6	3	241	1	0	6	9	0	76	3	0	1	0	1	0	1	0	0	0	0	1	0	0	1	1	0	0	2	7	59	631
Diadasia	0	1	23	1	0	0	0	0	3	0	0	0	42	0	0	0	0	0	61	0	0	5	0	4	0	0	0	0	0	0	0	1	0	0	0	0	0	1	69	211	
Dianthidium	0	0	10	0	86	1	0	0	6	0	0	0	22	0	0	0	0	0	25	0	0	0	0	0	0	1	0	0	0	0	0	3	0	0	0	0	0	0	27	181	
Diozya	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	
Dufourea	0	0	0	1	4	0	0	0	2	0	0	0	0	0	0	0	0	0	11	0	0	0	0	4	0	0	0	0	0	0	0	1	0	0	0	0	1	0	9	33	
Epeolus	0	1	4	0	78	1	0	0	1	0	0	0	13	0	2	0	0	0	3	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	1	0	1	107	
Epimelissodes	0	0	2	0	0	0	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	12	
Eucera	0	6	12	21	2	2	7	0	30	3	3	2	50	0	0	2	1	0	137	2	0	16	0	6	0	0	0	0	0	0	0	0	0	1	0	0	2	0	42	47	394
Habropoda	0	0	0	1	0	0	0	0	0	0	1	0	24	0	0	1	3	1	2	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	8	44
Halictus	0	3	23	42	482	823	2	4	48	2	18	0	162	0	3	35	118	3	146	26	0	28	0	52	0	5	0	1	12	7	13	18	4	10	0	22	48	24	179	2363	
Heriades	0	0	11	6	9	44	0	0	0	1	1	0	39	0	0	2	31	0	20	4	0	2	0	14	0	4	0	0	0	0	1	7	2	0	0	11	1	0	15	225	
Hoplitis	0	3	22	10	25	27	1	0	4	10	11	1	89	0	0	0	1	0	99	7	0	7	0	33	0	4	0	0	0	0	5	0	6	7	0	25	0	9	73	479	
Hylaeus	0	3	9	17	33	77	2	0	5	6	3	0	90	1	2	3	18	0	193	3	0	3	0	17	0	23	1	2	0	3	3	15	3	0	0	15	7	2	198	757	
Lasioglossum	4	5	16	44	288	601	1	2	26	0	3	9	169	0	2	33	47	1	316	47	2	24	2	21	2	14	4	3	36	10	24	9	6	159	0	12	6	68	319	2335	
Megachile	7	5	101	14	152	169	1	6	17	8	33	10	422	3	1	8	50	11	264	13	3	6	0	35	0	19	0	4	4	1	4	10	2	5	0	20	5	8	204	1625	
Melecta	0	0	0	1	1	0	1	0	0	0	0	1	16	0	0	0	0	0	8	0	0	1	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	1	8	41	
Melissodes	4	19	32	0	73	607	0	18	16	0	20	8	368	0	1	5	25	0	54	15	0	2	0	3	0	4	0	0	5	1	0	1	0	0	0	1	13	181	1476		
Nomada	0	4	9	10	31	68	6	0	13	4	1	1	39	0	0	3	21	0	81	6	0	27	0	15	0	14	0	0	5	0	6	10	6	21	0	14	1	52	44	512	
Nomia	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	23	0	0	0	25		
Osmia	0	7	49	92	134	96	1	3	37	38	22	13	499	0	0	41	62	5	410	64	0	25	7	116	0	29	4	3	10	15	7	45	1	7	0	67	18	45	419	2391	
Panurginus	0	0	1	16	55	26	1	0	1	0	0	2	0	0	0	13	4	1	40	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	13	21	196		
Perdita	0	0	11	13	0	0	8	0	7	0	10	0	63	0	0	0	0	1	0	28	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2	58	204	
Protosmia	0	0	0	1	0	6	0	0	0	0	0	0	1	0	0	0	0	1	2	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	16	
Pseudoanthidium	0	0	0	0	0	10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	
Sphecodes	0	0	1	4	85	17	0	1	7	2	1	1	23	0	0	13	18	0	50	5	0	1	0	1	0	10	0	0	2	2	2	1	1	23	0	6	1	2	41	321	
Stelis	0	0	0	1	0	9	1	0	1	1	0	0	7	0	0	0	0	0	9	1	0	0	0	1	0	1	2	0	0</												

Table 4: Number of bee specimens from each county, by species

	Adams	Asotin	Benton	Chelan	Chelan	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Inland	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Shagit	Shanawia	Snohomish	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whatcom	Whitman	Yakima	TOTAL	
<i>Agapostemon femoratus</i>	0	0	0	0	0	0	2	0	4	0	2	0	70	0	0	0	0	0	1	0	0	0	5	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	87
<i>Agapostemon subtilior</i>	0	0	1	0	31	142	0	2	25	1	0	0	35	0	1	10	15	0	13	1	0	0	0	4	0	0	0	1	1	0	6	0	0	14	0	1	1	1	6	312	
<i>Agapostemon virescens</i>	0	0	2	0	0	52	1	0	1	0	0	4	0	0	0	0	0	0	0	0	0	1	0	2	0	0	0	0	0	0	0	1	0	3	0	0	0	19	1	87	
<i>Andrena amphibola</i>	0	1	0	0	0	1	0	0	0	0	0	0	2	0	0	0	0	0	14	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	32	
<i>Andrena angustitarsata</i>	0	0	0	7	4	0	0	0	0	0	0	1	0	0	0	0	7	0	44	3	0	0	0	1	0	0	0	0	0	6	0	0	2	0	0	0	5	21	101		
<i>Andrena anisochlora</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Andrena astragali</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	5		
<i>Andrena auricoma</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	17	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	22		
<i>Andrena buckelli</i>	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	2	0	11	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	2	21	
<i>Andrena caerulea</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
<i>Andrena candida</i>	0	1	0	0	14	57	0	0	1	0	0	0	2	0	0	5	6	0	7	1	0	0	0	7	0	0	0	0	0	0	0	0	4	1	0	1	3	6	117		
<i>Andrena canothifloris</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
<i>Andrena chapmanae</i>	0	0	0	0	0	0	0	0	2	0	0	0	2	0	0	0	0	0	1	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	
<i>Andrena chlorogaster</i>	0	0	0	11	0	0	0	0	15	0	0	0	2	0	0	0	0	35	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	15	82		
<i>Andrena cledora</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Andrena colletina</i>	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
<i>Andrena columbiana</i>	0	0	0	0	18	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	22	
<i>Andrena commoda</i>	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	4		
<i>Andrena crataegi</i>	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	7	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	12		
<i>Andrena cressonii</i>	0	4	0	0	0	3	0	0	0	0	0	0	0	0	0	0	3	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	22		
<i>Andrena cupreincta</i>	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	5	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	14		
<i>Andrena cyanophila</i>	0	0	0	0	0	0	0	1	0	0	0	1	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	6		
<i>Andrena erythrogaster</i>	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Andrena forbesii</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1		
<i>Andrena frigida</i>	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	7	1	0	13		
<i>Andrena fuscicauda</i>	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	5	
<i>Andrena gordonii</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
<i>Andrena hemileuca</i>	0	0	0	0	4	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	8	
<i>Andrena hippotes</i>	0	0	0	1	0	5	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	14		
<i>Andrena illinoensis</i>	0	0	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6		
<i>Andrena knuthiana</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5		
<i>Andrena laurencei</i>	0	0	0	1	0	0	0	0	3	0	0	0	5	0	0	0	0	19	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	9	42		
<i>Andrena lupinorum</i>	0	0	0	1	0	1	0	0	0	0	0	3	1	0	0	0	0	16	1	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	31	
<i>Andrena macropuspinensis</i>	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2		
<i>Andrena mariae</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Andrena medianitens</i>	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	2	5		
<i>Andrena melanochroa</i>	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0	0	0	0	1	0	4	0	0	0	1	7	5	30			
<i>Andrena merriami</i>	0	0	0	3	0	0	0	0	1	0	0	0	4	0	0	0	0	45	4	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	19	82		
<i>Andrena microchlora</i>	0	0	0	3	0	0	0	0	3	0	0	13	9	0	0	0	0	91	1	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	42	168	
<i>Andrena milwaukeensis</i>	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
<i>Andrena miranda</i>	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	2	7	0	0	0	0	16	6	36		
<i>Andrena miserabilis</i>	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Andrena nigrithirta</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	5	6			
<i>Andrena nigrocervulea</i>	0	4	1	2	7	3	0	0	0	0	0	0	9	0	0	2	0	29	16	0	1	0	6	0	0	0	0	1	0	0	4	0	5	0	0	9	9	108			
<i>Andrena nivialis</i>	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7		
<i>Andrena rubecula</i>	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	4		
<i>Andrena obscuripostica</i>	0	0	0	0	3	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	4	12		

Table 4: Number of bee specimens from each county, by species (continued)

	Adams	Asotin	Benton	Chelan	Clallam	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Inland	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Shagit	Shanawia	Snohomish	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whatcom	Whitman	Yakima	TOTAL	
<i>Andrena vicinoides</i>	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	1	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	4	12	
<i>Andrena vulpicolor</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Andrena u-scripta</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	3		
<i>Andrena walleyi</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Andrena washingtoni</i>	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Anthidium robertsoni</i>	0	0	0	0	0	0	0	0	0	0	2	0	5	0	0	0	0	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	11	
<i>Anthidium atrifrons</i>	0	0	0	0	0	0	0	0	0	0	2	3	3	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	19	
<i>Anthidium banningsense</i>	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	7	
<i>Anthidium clypeodentatum</i>	0	0	0	0	0	0	0	3	0	0	0	4	0	0	0	0	0	0	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	
<i>Anthidium emarginatum</i>	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	5
<i>Anthidium formosum</i>	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2	
<i>Anthidium manicatum</i>	0	0	2	0	4	9	0	0	0	0	1	0	1	1	0	1	4	4	0	0	0	0	0	0	0	0	0	1	0	0	1	3	0	1	0	0	0	0	0	1	34
<i>Anthidium mormonum</i>	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1	0	0	0	4	0	0	0	0	0	0	0	2	1	0	0	0	0	0	4	15	
<i>Anthidium oblongatum</i>	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	23	
<i>Anthidium tenuiflorae</i>	0	0	0	1	3	0	0	0	0	0	0	0	1	0	0	0	0	0	7	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	4	18
<i>Anthidium utahense</i>	0	0	0	0	0	0	0	0	1	1	0	0	5	0	0	0	0	0	4	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	17
<i>Anthophora affabilis</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Anthophora albata</i>	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Anthophora bomboides</i>	0	0	0	0	27	0	0	0	0	0	0	6	0	0	24	0	0	5	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2	1	8	1	76	
<i>Anthophora crotchii</i>	0	0	6	0	0	0	0	0	0	0	0	3	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	18	
<i>Anthophora curta</i>	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	
<i>Anthophora edwardsi</i>	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	
<i>Anthophora pacifica</i>	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	
<i>Anthophora terminalis</i>	0	0	0	0	7	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	2	0	0	9	21	
<i>Anthophora urbana</i>	0	0	0	0	0	1	0	0	0	0	9	0	20	0	0	0	0	0	7	3	0	0	0	14	0	0	0	0	1	0	0	1	0	0	0	0	3	0	6	65	
<i>Anthophora ursina</i>	0	0	2	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	9	
<i>Apis mellifera</i>	0	0	2	6	12	59	1	0	0	0	10	0	58	0	22	2	6	5	31	4	0	3	9	0	9	0	1	0	1	7	4	0	9	3	6	0	1	0	1	24	287
<i>Ashmeadiella aridula</i>	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	
<i>Ashmeadiella buconis</i>	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	6	
<i>Ashmeadiella cactorum</i>	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4
<i>Ashmeadiella californica</i>	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Ashmeadiella cubiceps</i>	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3
<i>Ashmeadiella difugata</i>	0	0	0	1	0	0	0	0	0	0	0	7	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	13	
<i>Ashmeadiella fozzella</i>	0	3	0	1	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	15	
<i>Ashmeadiella meliloti</i>	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	
<i>Ashmeadiella sculleni</i>	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Ashmeadiella timberlakei</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Atoposmia abjecta</i>	0	0	0	0	0	0	0	0	0	0	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	1	7	
<i>Atoposmia copelandica</i>	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	7	15
<i>Atoposmia elongata</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	2	
<i>Bastes fulviventris</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	14	
<i>Bombus appostus</i>	0	1	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	
<i>Bombus caliginosus</i>	0	0	0	0	1	10	0	0	0	0	0	0	3	0	26	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	44	
<i>Bombus centralis</i>	0	1	0	1	0	0	0	5	0	0	0	19	0	0	0	0	0	18	1	0	3	0	7	0	0	0	0	0	0	0	0	1	1	0	0	5	0	8	31	101	
<i>Bombus fervidus</i>	0	0	0	0	3	10	2	0	2	0	0	5	4	0	0	0	1	0	3	4	0	0	6	0	0	0	0	0	0	0	0	0	0	2	0	1	0	2	12	57	
<i>Bombus flavidus</i>	0	0	0	3	20	0	0	0	0	0	0	0	0	0	0	6	0	1	0	0	0	0	1	0	3	0	0	0	1	1	1	0	0	0	0	0	0	0	2		

Table 4: Number of bee specimens from each county, by species (continued)

	Adams	Asotin	Benton	Chelan	Clallam	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Inland	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Shagit	Shanawia	Snohomish	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whatcom	Whitman	Yakima	TOTAL	
<i>Ceratina pacifica</i>	0	0	0	0	0	0	0	0	1	0	0	0	16	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	9	28	
<i>Chelostoma minutum</i>	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	7	
<i>Coelioxys alternatus</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	3	
<i>Coelioxys apachorum</i>	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4
<i>Coelioxys banksi</i>	0	0	0	0	4	2	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	
<i>Coelioxys edita</i>	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Coelioxys funeraria</i>	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	5	
<i>Coelioxys gilensis</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	2	
<i>Coelioxys grindeliae</i>	0	0	1	0	0	0	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	11	
<i>Coelioxys moestus</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Coelioxys octodentata</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	
<i>Coelioxys octodentatus</i>	0	1	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	
<i>Coelioxys rufitarsis</i>	0	1	0	0	14	10	0	0	0	0	0	0	4	0	0	0	9	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	1	41
<i>Coelioxys serriocaudatus</i>	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Coelioxys sodalis</i>	0	0	0	0	1	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	2	0	0	0	0	0	1	7
<i>Colletes compactus</i>	0	0	1	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	5	
<i>Colletes consors</i>	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	13	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	25
<i>Colletes fulgidus</i>	0	3	0	1	7	1	0	0	3	0	0	1	7	1	0	0	4	0	17	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1	3	5	56
<i>Colletes gypsicolens</i>	0	2	11	0	0	0	0	0	1	0	2	1	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	8	35	
<i>Colletes inaequalis</i>	0	2	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	8
<i>Colletes kincaidii</i>	0	1	0	0	99	0	0	0	1	0	0	0	0	0	0	6	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	4	114
<i>Colletes lutzii</i>	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9
<i>Colletes nigrifrons</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	
<i>Colletes phacellae</i>	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	
<i>Colletes simulans</i>	0	0	1	0	0	0	0	0	2	0	1	0	10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	16
<i>Colletes slevini</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	9
<i>Diadasia australis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2	
<i>Diadasia diminuta</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	7
<i>Diadasia enavata</i>	0	1	0	0	0	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	13	
<i>Diadasia lutzii</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Diadasia nitidifrons</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	3	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	18	26
<i>Diadasia ochracea</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	41	41
<i>Dianthidium curvatum</i>	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
<i>Dianthidium parvum</i>	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
<i>Dianthidium pudicum</i>	0	0	0	0	0	0	0	1	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	9
<i>Dianthidium subparvum</i>	0	0	0	0	68	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	72
<i>Dianthidium ulkei</i>	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	6	0	0	0	0	0	0	1	0	0	0	0	0	2	0	0	0	0	0	0	0	0	3	13
<i>Diozys aurifuscus</i>	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
<i>Diozys pomonae</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
<i>Dufourea dilatipes</i>	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6
<i>Dufourea maura</i>	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	2
<i>Dufourea trochantera</i>	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	4	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	6	13
<i>Epeolus americanus</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	2
<i>Epeolus emiliae</i>	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
<i>Epeolus minimus</i>	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	4
<i>Epeolus olympiellus</i>	0	0	0	0	76	0	0	0	0	0	0	0	1	0	2	0	0	0	0	0																					

Table 4: Number of bee specimens from each county, by species (continued)

	Adams	Asotin	Benton	Chelan	Clallam	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Inland	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Shagit	Shanawia	Snohomish	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whatcom	Whitman	Yakima	TOTAL		
<i>Lasiosglossum titusi</i>	0	0	0	1	9	137	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	0	0	0	14	0	168		
<i>Lasiosglossum trizonatum</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	6		
<i>Lasiosglossum villosulum</i>	0	0	0	0	7	65	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	7	0	3	0	0	20	0	0	0	0	1	105		
<i>Lasiosglossum yukonae</i>	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6		
<i>Lasiosglossum zephyrum</i>	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2	0	7		
<i>Lasiosglossum zonulum</i>	0	0	0	0	13	3	0	0	0	0	0	0	0	0	0	1	3	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	4	0	1	0	0	0	27		
<i>Lasiosglossum zonulus</i>	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4		
<i>Megachile angelarum</i>	0	0	2	0	5	30	0	0	0	0	0	2	0	9	0	1	7	2	4	0	0	0	0	1	0	2	0	0	0	0	3	0	1	0	0	0	0	0	0	0	5	74
<i>Megachile apicalis</i>	0	0	3	0	0	0	1	0	1	0	6	0	23	0	0	0	0	0	11	1	0	1	0	2	0	0	0	0	0	1	0	1	1	0	0	11	0	4	10	77		
<i>Megachile brevis</i>	0	1	2	1	3	6	0	0	0	0	0	0	10	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	31		
<i>Megachile centuncularis</i>	0	0	0	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4		
<i>Megachile coquillettii</i>	0	0	0	0	0	0	0	0	1	0	0	0	6	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	11	
<i>Megachile fidelis</i>	0	0	4	0	1	8	0	1	0	0	0	0	0	0	0	4	4	1	5	5	0	0	0	1	0	2	0	0	0	0	0	0	0	0	1	0	0	0	0	6	43	
<i>Megachile frigida</i>	0	0	0	0	15	4	0	0	0	0	0	0	0	0	0	0	3	0	1	0	0	0	0	4	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	29	
<i>Megachile gemula</i>	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	6	
<i>Megachile gravata</i>	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Megachile lapponica</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	2	4
<i>Megachile lippiae</i>	0	0	2	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	9	
<i>Megachile melanophaea</i>	0	0	0	0	7	0	0	0	0	1	0	1	0	0	0	1	3	0	3	0	0	0	0	4	0	0	0	1	1	0	0	0	0	0	4	0	0	1	0	1	28	
<i>Megachile mellitarsis</i>	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Megachile mendica</i>	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Megachile montivaga</i>	0	0	1	1	12	9	0	0	1	0	1	0	1	0	0	0	1	0	0	0	1	0	0	2	0	5	0	0	0	0	0	0	0	0	0	0	3	0	1	1	40	
<i>Megachile nevadensis</i>	2	0	15	1	0	0	0	0	3	0	10	0	43	0	0	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	16	110	
<i>Megachile onobrychidis</i>	0	0	3	0	0	0	0	0	3	1	5	0	67	0	0	0	0	0	14	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	16	113	
<i>Megachile parallela</i>	0	0	0	0	0	0	0	0	0	0	0	0	10	0	0	0	0	0	2	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	15	
<i>Megachile perthirta</i>	0	4	3	2	70	76	0	3	1	4	1	9	23	3	1	2	24	8	15	2	0	1	0	4	0	2	0	3	2	0	0	5	0	0	0	1	3	3	17	292		
<i>Megachile pugnata</i>	0	0	3	3	13	2	0	1	1	0	2	0	0	0	0	0	0	0	6	2	2	0	0	4	0	2	0	0	0	0	0	0	0	0	0	1	0	0	10	52		
<i>Megachile relativa</i>	0	0	0	0	13	0	0	0	0	1	0	0	0	0	0	0	1	0	16	0	0	0	0	2	0	4	0	0	0	0	1	0	0	0	0	1	0	0	1	0	2	41
<i>Megachile rotundata</i>	0	0	11	0	0	19	0	1	0	1	5	0	28	0	0	0	6	0	3	0	0	0	0	2	0	0	0	0	0	0	0	1	0	0	0	0	3	0	0	3	83	
<i>Megachile subnigra</i>	0	0	1	0	0	0	0	0	2	0	0	0	6	0	0	0	0	0	11	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	31	
<i>Megachile tezana</i>	0	0	0	0	0	4	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	17	
<i>Megachile umatillensis</i>	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Megachile wheeleri</i>	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	12	
<i>Melecta pacifica</i>	0	0	0	0	0	0	1	0	0	0	0	0	1	12	0	0	0	0	6	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	24
<i>Melecta separata</i>	0	0	0	1	0	0	0	0	0	0	0	0	3	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	6	13		
<i>Melecta thoracica</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	
<i>Melissodes agilis</i>	0	13	2	0	0	0	2	0	0	2	0	0	5	61	0	0	0	0	0	3	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	9	9	108	
<i>Melissodes binatrix</i>	0	4	12	0	0	0	0	0	0	0	6	3	37	0	0	0	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	9	75		
<i>Melissodes communis</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Melissodes dagosus</i>	0	0	0	0	0	0	0	0	0	0	0	0	24	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	25	
<i>Melissodes glenwoodensis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Melissodes lupinus</i>	0	0	0	0	0	48	0	0	0	0	0	0	1	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	62		
<i>Melissodes lustrus</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	101	107		
<i>Melissodes menuechus</i>	0	0	0	0	0	0	0	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9		
<i>Melissodes metenus</i>	0	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	11		
<i>Melissodes microstictus</i>	0	1	0	0	41	62	0</																																			

	Adams	Asotin	Benton	Chelan	Columbia	Clark	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Island	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Skagit	Skamania	Snohomish	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whitcom	Whitman	Yakima	TOTAL	
<i>Osmia cobaltina</i>	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	
<i>Osmia coloradensis</i>	0	0	0	9	1	0	0	0	1	1	0	1	0	0	1	1	2	11	0	0	1	0	7	0	4	0	2	1	0	0	3	0	0	0	0	1	1	3	15	67
<i>Osmia cornifrons</i>	0	0	0	0	0	6	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	
<i>Osmia cyanella</i>	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	
<i>Osmia densa</i>	0	0	0	5	1	0	0	0	0	1	0	0	0	0	0	0	0	8	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	23
<i>Osmia ednae</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6
<i>Osmia exigua</i>	0	0	1	2	1	1	0	0	0	0	2	3	0	0	2	0	0	10	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	11	35	
<i>Osmia giljarum</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia grinnelli</i>	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	24
<i>Osmia integro</i>	0	0	0	0	0	0	0	0	0	4	0	29	0	0	0	0	0	8	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	10	58	
<i>Osmia inurbana</i>	0	0	0	0	0	0	0	0	1	1	2	0	1	0	0	0	0	5	2	0	1	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6	25	
<i>Osmia iridis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia juxta</i>	0	0	0	2	6	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	
<i>Osmia lacta</i>	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia lignaria</i>	0	0	0	1	4	6	0	0	0	0	0	3	0	0	3	2	0	4	1	0	0	3	0	0	0	0	0	0	6	1	0	0	0	0	0	1	0	5	40	
<i>Osmia longula</i>	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	7		
<i>Osmia malina</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia montana</i>	0	0	0	10	0	0	0	0	0	0	1	2	0	0	0	0	0	22	9	0	1	0	27	0	2	0	0	0	0	0	0	0	0	0	0	0	1	26	101	
<i>Osmia nemoris</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Osmia nifoata</i>	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	5		
<i>Osmia nigriventris</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia odontogaster</i>	0	0	0	1	4	0	0	0	0	0	3	0	0	0	0	0	16	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	32	
<i>Osmia pentstemonis</i>	0	0	0	11	1	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	3	0	4	4	0	1	1	0	0	0	0	0	0	0	0	0	0	4	33	
<i>Osmia phacellae</i>	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	8	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	
<i>Osmia proxima</i>	0	0	0	0	2	2	0	0	0	0	0	0	0	0	1	9	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	1	1	19		
<i>Osmia pusilla</i>	0	0	0	5	5	10	0	0	0	0	0	0	0	0	0	0	27	0	0	0	0	3	0	1	0	0	0	0	0	0	0	0	0	6	0	1	1	59		
<i>Osmia rawlini</i>	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	7		
<i>Osmia regulina</i>	0	0	0	1	0	0	0	0	0	5	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13	0	0	2	25		
<i>Osmia sculleni</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6		
<i>Osmia sedula</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	
<i>Osmia subaustriis</i>	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	6	0	0	0	4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	4	19		
<i>Osmia tezara</i>	0	0	0	0	0	4	0	0	0	0	0	1	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	11	
<i>Osmia trifoliaria</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1		
<i>Osmia tristella</i>	0	0	0	7	6	0	0	0	0	0	1	0	0	0	0	5	0	17	0	0	0	1	0	16	0	0	2	0	0	0	0	0	0	0	1	0	30	86		
<i>Osmia vandykei</i>	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	15	
<i>Panurginus atriceps</i>	0	0	0	11	45	9	0	0	0	0	2	0	0	10	1	0	35	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	8	4	126		
<i>Panurginus inepsus</i>	0	0	0	5	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7		
<i>Perdita claypolei</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	11		
<i>Perdita lingualis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2			
<i>Perdita nevadensis</i>	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	24		
<i>Perdita nuda</i>	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
<i>Perdita oregonensis</i>	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5		
<i>Perdita salicis</i>	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
<i>Perdita wyomingensis</i>	0	0	0	1	0	0	8	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	13		
<i>Perdita zonalis</i>	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2		
<i>Protosmia rubifrons</i>	0	0	0	1	0	6	0	0	0	0	0	1	0	0	0	0	1	2	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	16	
<i>Pseudanthidium nanum</i>	0	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9		
<i>Sphecodes pecosensis</i>	0	0	0	0	3	5	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	2	16		
<i>Stelis callura</i>	0	0	0	0	0	0	0	0	0																															

Table 5: Your determination accuracy in 2024.

Taxon
No specimens identified

7 Taxonomic Accuracy, 2024

In 2024, you identified 0 of your 43 specimens to genus level and 0 to species level (see Table 5). In total, volunteers from the Washington Bee Atlas project identified 1.3 % (127) of the 10006 bee specimens to the level of genus, with an average accuracy of 87.4%. Volunteer identifications to species level did not occur or are not included in the data (see Table 6).

Table 6: Determination accuracy for all volunteers in 2024.

Taxon	Specimens ID-ed	Correct ID	% Correct
Family			
<i>Andrenidae</i>	21	16	76.2
<i>Apidae</i>	52	52	100.0
<i>Colletidae</i>	6	3	50.0
<i>Halictidae</i>	8	6	75.0
<i>Megachilidae</i>	43	41	95.3
<i>TOTAL</i>	130	118	90.8
Genus			
<i>Agapostemon</i>	1	1	100.0
<i>Andrena</i>	5	0	0.0
<i>Anthidium</i>	1	1	100.0
<i>Anthophora</i>	6	6	100.0
<i>Ceratina</i>	7	7	100.0
<i>Coelioxys</i>	2	2	100.0
<i>Colletes</i>	4	3	75.0
<i>Eucera</i>	1	1	100.0
<i>Halictus</i>	2	0	0.0
<i>Hoplitis</i>	4	4	100.0
<i>Hylaeus</i>	2	0	0.0
<i>Lasioglossum</i>	5	4	80.0
<i>Megachile</i>	18	18	100.0
<i>Melissodes</i>	36	33	91.7
<i>Osmia</i>	17	15	88.2
<i>Perdita</i>	16	16	100.0
<i>TOTAL</i>	127	111	87.4
Species			
<i>TOTAL</i>	0	0	NaN

Table 7: Your determination accuracy.

Taxon
No specimens identified

8 Taxonomic Accuracy, All Years

Over your time in the Atlas you identified 0 of your 113 specimens to genus level and 0 to species level (see Table 7). In total, volunteers from the Washington Bee Atlas project identified 0.6 % (127) of the 22418 bee specimens to the level of genus, with an average accuracy of 87.4%. Volunteer identifications to species level did not occur or are not included in the data (See Table 8).

Table 8: Determination accuracy for all volunteers.

Taxon	Specimens ID-ed	Correct ID	% Correct
Family			
<i>Andrenidae</i>	21	16	76.2
<i>Apidae</i>	52	52	100.0
<i>Colletidae</i>	6	3	50.0
<i>Halictidae</i>	8	6	75.0
<i>Megachilidae</i>	43	41	95.3
<i>TOTAL</i>	130	118	90.8
Genus			
<i>Agapostemon</i>	1	1	100.0
<i>Andrena</i>	5	0	0.0
<i>Anthidium</i>	1	1	100.0
<i>Anthophora</i>	6	6	100.0
<i>Ceratina</i>	7	7	100.0
<i>Coelioxys</i>	2	2	100.0
<i>Colletes</i>	4	3	75.0
<i>Eucera</i>	1	1	100.0
<i>Halictus</i>	2	0	0.0
<i>Hoplitis</i>	4	4	100.0
<i>Hylaeus</i>	2	0	0.0
<i>Lasioglossum</i>	5	4	80.0
<i>Megachile</i>	18	18	100.0
<i>Melissodes</i>	36	33	91.7
<i>Osmia</i>	17	15	88.2
<i>Perdita</i>	16	16	100.0
<i>TOTAL</i>	127	111	87.4
Species			
<i>TOTAL</i>	0	0	NaN